

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Project Name: SIM Crime Reporting & Mapping System

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Course Code: AAUA-SEN202

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the SIM Crime Reporting & Mapping System. This front-end application provides an easy-to-use portal for individuals to log campus incidents and view them instantly on a dynamic visual map grid.

1.2 Scope

The application is a client-side prototype developed to demonstrate interactive crime reporting capabilities. The system tracks local security issues such as petty theft, bullying, physical assaults, and other threats to community safety.

1.3 Intended Audience

This specification is intended for the course evaluator, Dr. O. O. Ajayi, student, Staff, Landlord and Security Personnel.

2. Overall Description

2.1 Product Perspective

The SIM system functions entirely as an independent frontend application running inside modern mobile and desktop web browsers. It requires no active backend installations or localized database configurations to demonstrate its behavior during evaluation.

2.2 User Classes

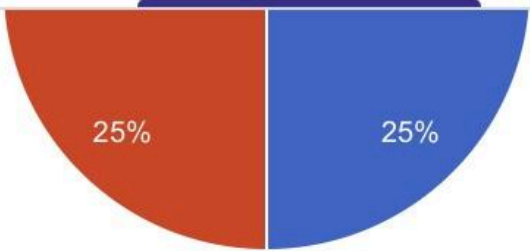
- **Community Resident & Landlord:** Includes students, campus staff, and local landlords who use the web form to log details and check the map canvas.
- **Security Operator:** Campus safety officers who analyze the live map and review newly filed entries in the recent history log.

3. Survey & Data Summary

3.1 Survey Findings

Untitled form

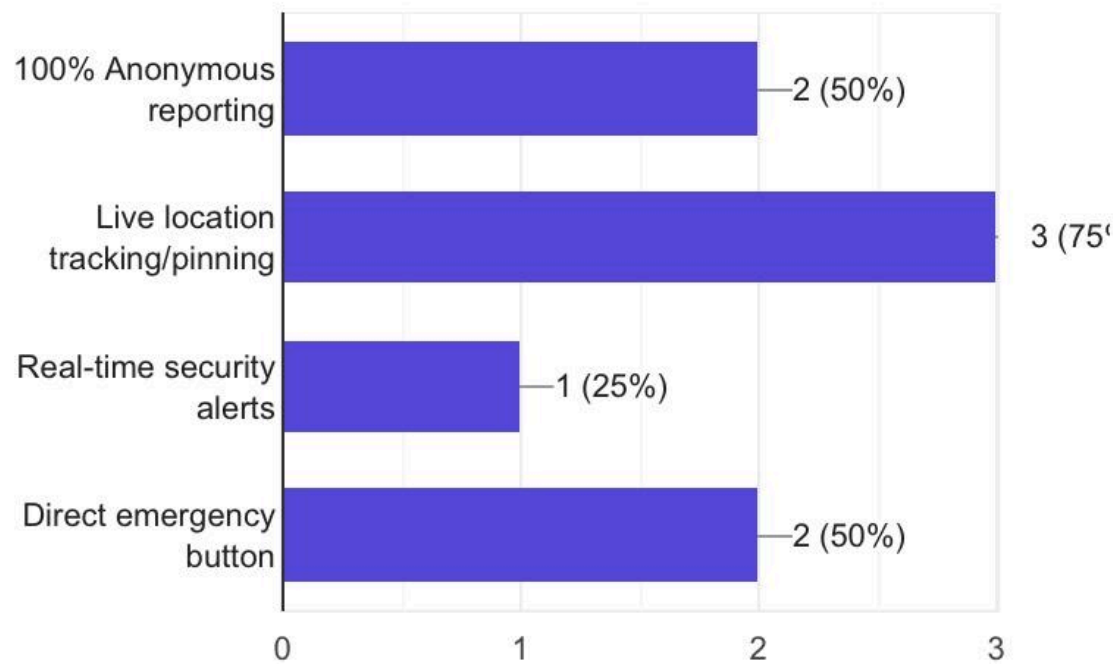
Questions Responses **4** Settings



What feature would make you use a digital crime-reporting app?

 Copy chart

4 responses



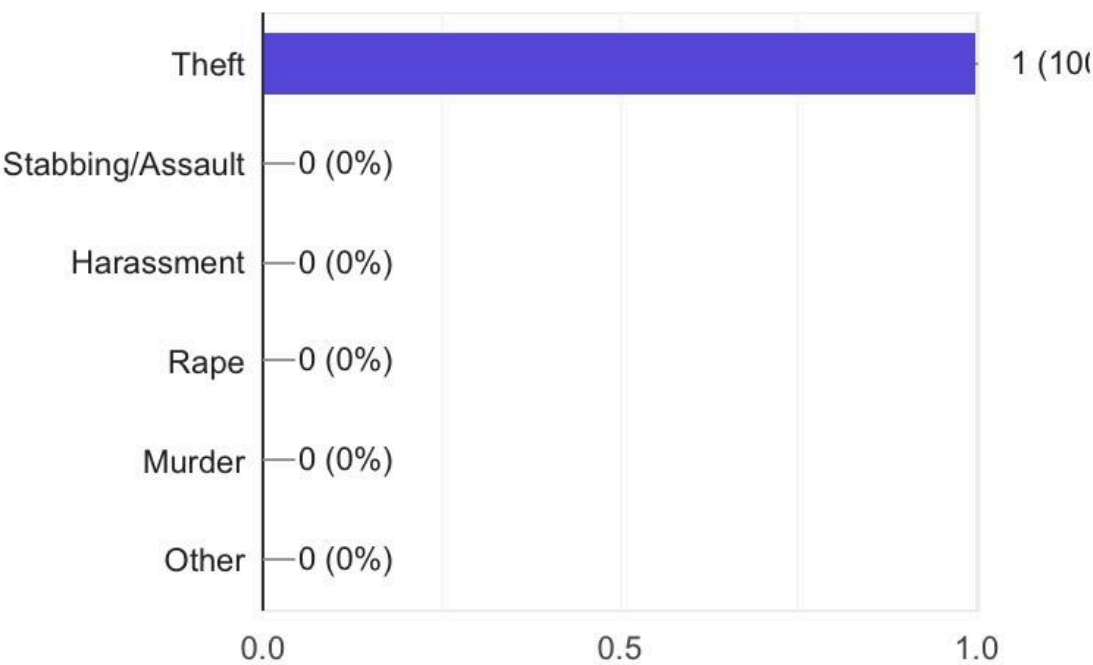
Untitled form

Questions **Responses 4** Settings

If yes, what type of incident?

 [Copy chart](#)

1 response



Did you report the incident?

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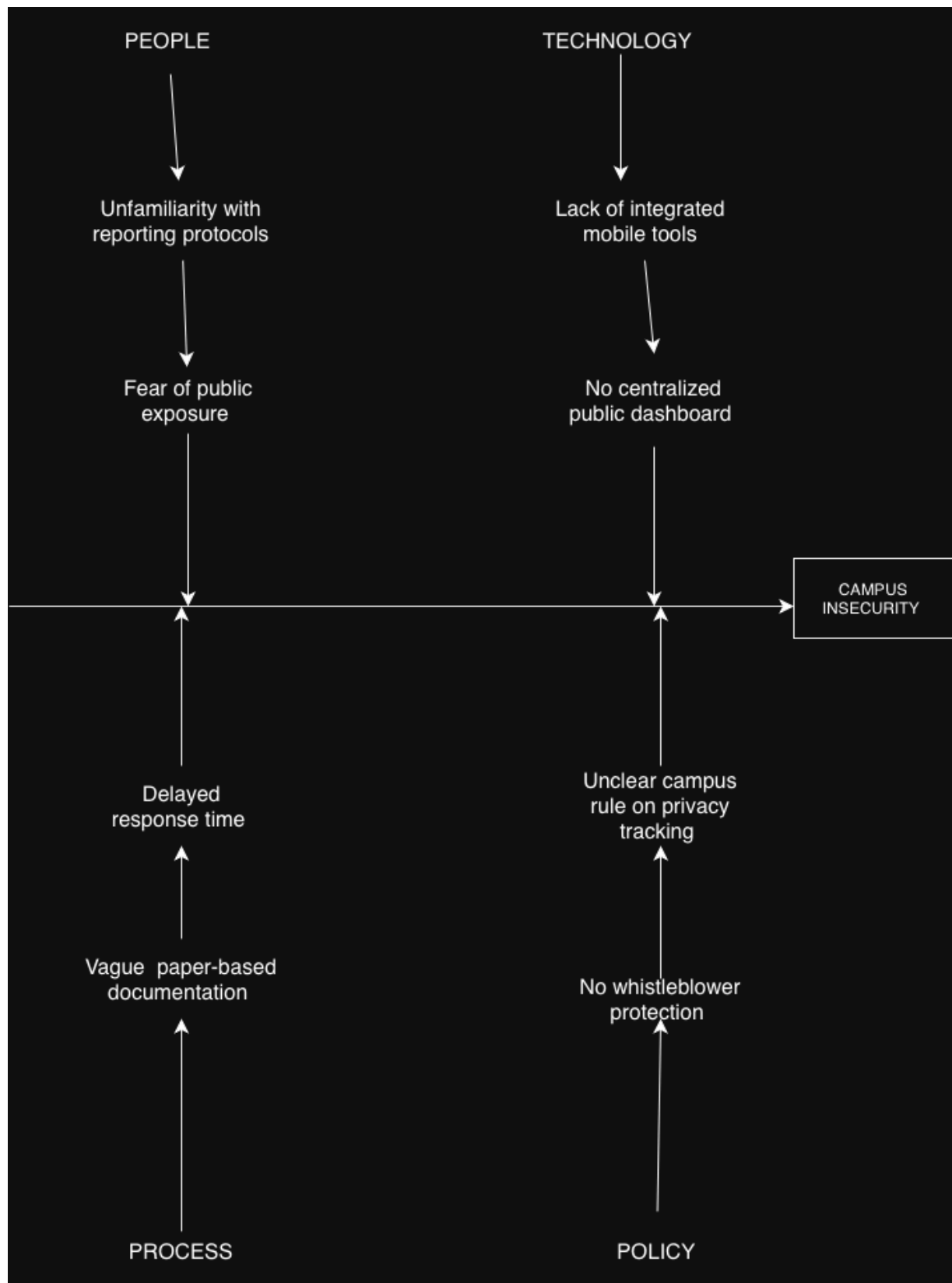
4 responses



Our quantitative Google Form data shows a critical need for an online system:

- **Demographics:** 100% of the polled participants are currently university students.
- **Incident History:** 25% of respondents have witnessed or personally experienced a security issue on or around campus.
- **Incident Nature:** 100% of those experienced incidents involved petty theft or stealing.
- **Reporting Hesitancy:** Among those who stayed silent, 50% stated they didn't believe any official action would be taken by authorities.
- **Desired Functions:** 66.7% requested a direct alert feature, while 33.3% prioritized an interactive incident map, photo uploads, and report tracking capabilities.

3.2 System Root Cause Analysis



Our Ishikawa fishbone diagram highlights several distinct flaws in current security procedures:

- **People:** General fear of public exposure, fear of retaliation, and an unfamiliarity with reporting protocols.
- **Process:** Sluggish response times from security guards due to clunky, paper-bound incident logging.
- **Technology:** A complete lack of dedicated reporting software, no shared safety dashboards, and zero visual map tracking capabilities.

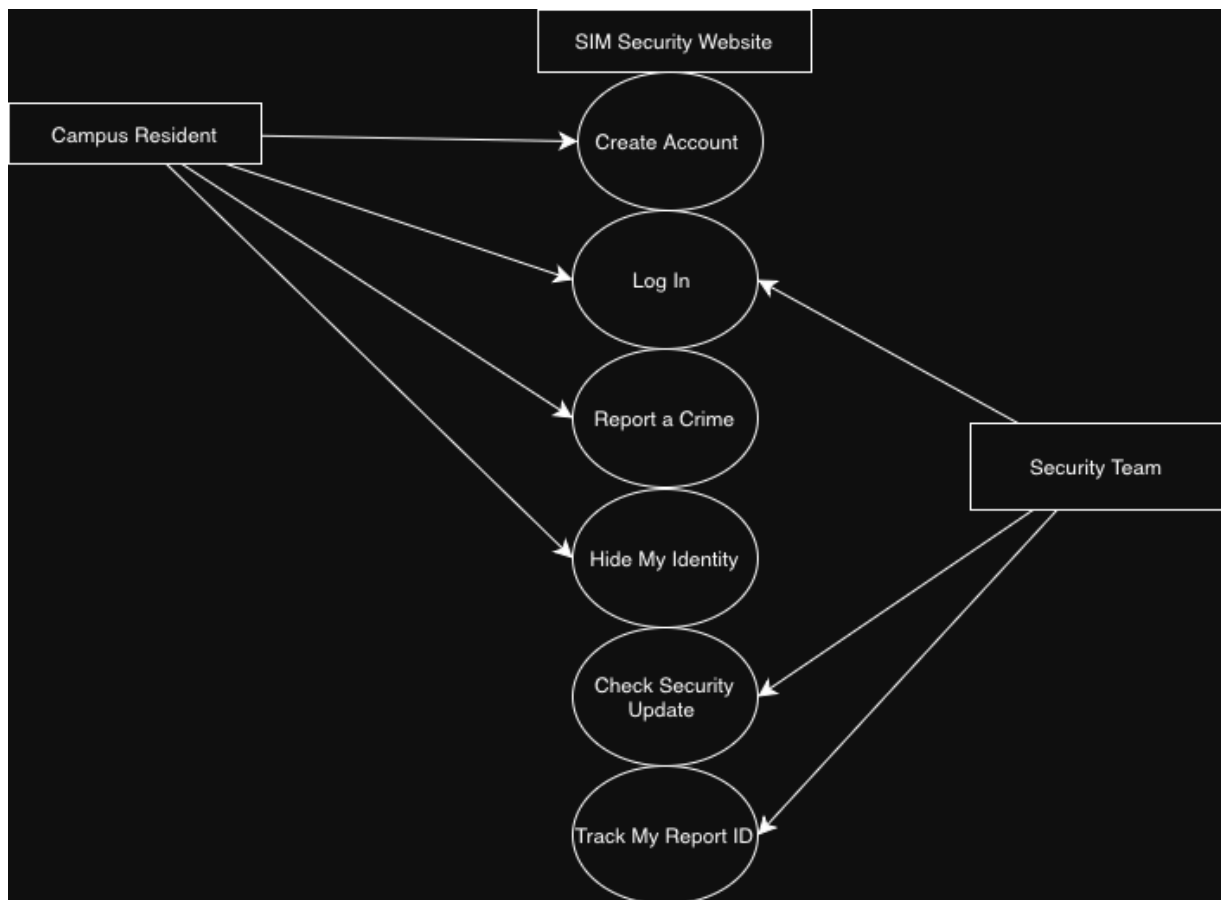
4. Functional Requirements

- **FR1:** The form interface must capture user metrics including role selection, incident category, textual description, and specific location text.
- **FR2:** The system must render the anonymity toggle checkbox directly below the department field input box.
- **FR3:** When the anonymity option is toggled active, the system shall dynamically hide the name and department inputs from the form panel area.
- **FR4:** Upon form submission, the code must generate a unique, non-repeating tracking string adhering to the format code: **SIM-2026-XXXX**.
- **FR5:** The system must instantly write the submission data as a new record line at the top of the local web ledger table.
- **FR6:** The interface shall dynamically create and append a new glowing radar map beacon within the boundaries of the map display frame upon every submission.

5. Non-Functional Requirements

- **NFR1 (Usability):** The application interface shall be intuitive enough for non-technical community users to operate without prior instructions.
- **NFR2 (Availability):** The application must remain fully functional and responsive across standard mobile and desktop viewports without structural degradation.
- **NFR3 (Security/Privacy):** Reports submitted anonymously must completely exclude personal identification data from the browser's viewable table rows.
- **NFR4 (Performance):** Form validation, confirmation page rendering, and map point plotting must complete execution within 2 seconds of clicking submit.

6. Use-Case Diagram Analysis



Our system diagram establishes clear boundaries between system interactions:

- **Community Resident & Landlord Role:** This actor interacts with the system to log a new incident report, select anonymous masking, upload image strings, and track their verification codes.
- **Security Operator Role:** This administrative actor interacts with the interface to view the comprehensive submitted report records, monitor map beacon clusters, and manage recent entries.

7. Form Fields & Elements (Object Listing)

UI Element ID	Element Type	Input Constraints & Validation	Action Handler Behavior
userClass	Select Dropdown	Required. Options: Student, Staff, Landlord.	Captures and classifies the role of the reporting entity.

infractionType	Select Dropdown	Required. Options: Stealing, Bullying, Assault, Sexual Assault, Murder.	Determines the classification category for the report logging.
sceneLocation	Text Input Field	Required. String length 1–100 characters.	Stores the physical description or campus zone name.
narrativeDetails	Text Area Box	Optional. Plain text string.	Logs specific details regarding the nature of the event.
reporterName	Text Input Field	Optional. Hidden if anonCheck is true.	Records the full legal name of the person reporting.
reporterDept	Text Input Field	Optional. Hidden if anonCheck is true.	Logs the academic or staff department division.
anonCheck	Input Checkbox	Optional. Boolean (True/False).	Toggles the active display state of identity fields on change.
dispatchBtn	Action Button	Interactive on-click listener execution.	Triggers form validation, token generation, and map plotting.

8. System Architecture

The SIM prototype is constructed utilizing an agile, three-tier frontend client architecture:

- **Markup Layer (HTML5):** Defines the structural skeleton, container boxes, tables, and dropdown form elements.
- **Presentation Layer (CSS3):** Applies the modern layout rules, responsive design queries, and real-time radar pulsing animation keyframes.
- **Logical Engine (JavaScript):** Coordinates DOM selection handlers, input validation, unique string generation, and absolute coordinate plotting algorithms.

The system uses local browser session states to maintain reports live on screen. For full deployment, this interface would link directly to a relational cloud database platform to handle persistent cross-user storage.

9. Conclusion

The SIM Crime Reporting & Mapping System introduces a modern, web-based alternative to outdated paper incident logs. By eliminating complex interfaces and giving users full control over their personal privacy, the application directly counters the main reporting obstacles identified in our campus data survey. This frontend prototype demonstrates how smart interactive maps can improve crime data visibility and boost overall public safety awareness.